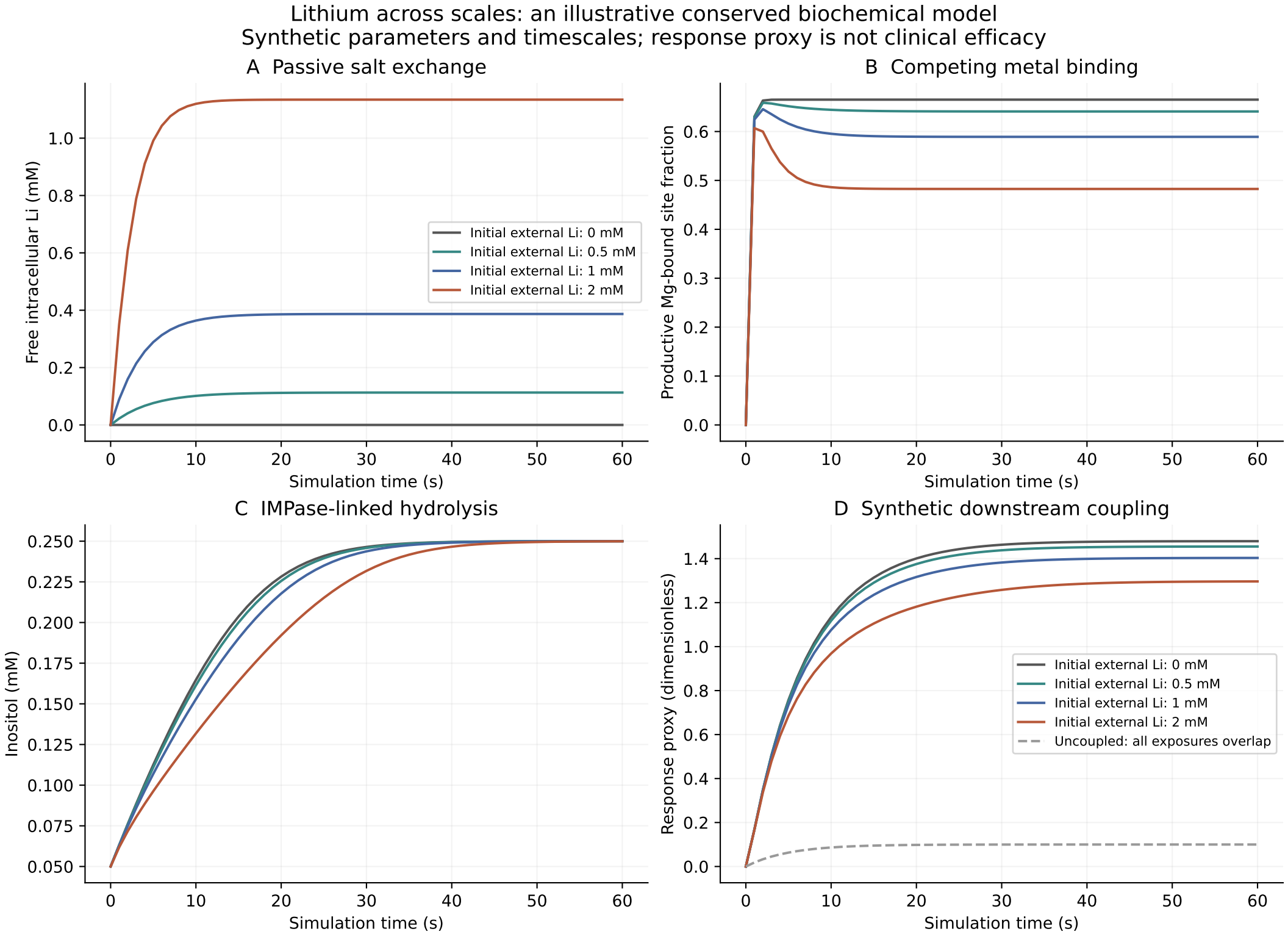
Lithium across scales: an illustrative conserved biochemical model
Synthetic parameters and timescales; response proxy is not clinical efficacy
A Passive salt exchange
Free intracellular Li (mM)
Simulation time (s)
Initial external Li: 0 mM
Initial external Li: 0.5 mM
Initial external Li: 1 mM
Initial external Li: 2 mM
B Competing metal binding
Productive Mg-bound site fraction
Simulation time (s)
C IMPase-linked hydrolysis
Inositol (mM)
Simulation time (s)
D Synthetic downstream coupling
Response proxy (dimensionless)
Simulation time (s)
Initial external Li: 0 mM
Initial external Li: 0.5 mM
Initial external Li: 1 mM
Initial external Li: 2 mM
Uncoupled: all exposures overlap